

Sanctions without Markups: Sourcing as the Margin of Bureaucratic Inefficiency*

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Abstract

Do sanctioned officials face higher prices because suppliers extract delivery-risk premia, or because the buyer’s sourcing pattern shifts? The two predictions are observationally equivalent in cross-buyer data. Using bid-level data on 479,330 pharmaceutical purchases by the São Paulo State Department of Health (2009–2019), we exploit a parallel administrative-request channel that shares all planning constraints of court-mandated procurement but carries no penalty for officials. Manski-Lee bounds place the selection-corrected “under-the-gun” gap at [15.9%, 21.1%]. Within firm-buyer-item triples observed under both regimes, the Admin coefficient is statistically indistinguishable from zero ($\hat{\beta} = 0.035$, $SE = 0.041$): *there is no within-firm markup*. The price margin operates through demand fragmentation—admin orders are roughly $3.3\times$ larger, mechanically delivering bulk discounts—and through equilib-

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rium changes in which firm wins. The bounded UTG implies a per-unit-price welfare cost of \$27.8 M per year on the \$300 M of annual litigated spending in São Paulo; the policy lever is demand aggregation, not contract design.

Keywords: public procurement, health litigation, enforcement costs, judicial mandates, accountability design, sourcing

JEL: D44, D73, H51, H57, I18, K41

1. Introduction

Do sanctioned officials face higher prices because suppliers extract delivery-risk premia, or because the buyer’s sourcing pattern shifts? Standard models of bureaucratic accountability ([Prendergast, 2007](#)) predict the first; passive-waste accounts ([Bandiera et al., 2009](#)) predict the second. The two predictions are observationally equivalent in cross-buyer data, where buyer-supplier match heterogeneity confounds within-firm and between-firm margins. We provide a within firm-buyer-item test in a setting where accountability pressure should bite hardest: court-mandated pharmaceutical procurement in São Paulo under fines, personal liability, and fund seizure for delivery failure. The within-firm prediction fails; the sourcing prediction holds.

Setting and identification asset.. We use all pharmaceutical procurement by the São Paulo State Department of Health (SES/SP) from 2009 through 2019: 479,330 purchase-offer-item observations on the state’s electronic procurement platform (BEC), which is the mandatory channel for common goods since 2007. The institutional asset that identifies our comparison is a parallel

administrative-request channel, instituted by the SES/SP scientific committee in 2009: it shares all planning constraints of court-mandated procurement (timelines of 1–10 days, small quantities, diverted budget, identical auction procedures on BEC) but carries no penalty for officials when delivery fails. To our knowledge, no other Brazilian state operates a comparable dual channel for urgent pharmaceutical demand.

Findings.. First, there is no within-firm markup. Within firm-buyer-item triples observed under both regimes (1,206 triples, 4,573 observations), the Admin coefficient is $\hat{\beta} = 0.035$ (SE 0.041), statistically indistinguishable from zero. The same firm prices the same item to the same buyer at the same level under either regime; the delivery-risk-premium prediction is empirically rejected. *Second, the bounded under-the-gun gap is modest.* Under Manski-Lee monotone restrictions on committee selection, the gap lies in [15.9%, 21.1%], at the lower end of within-buyer procurement-cost dispersion documented in comparable settings (e.g., [Best et al., 2023](#); [Bosio et al., 2022](#)). *Third, the gap operates through demand fragmentation.* Administrative orders are roughly $3.3\times$ larger than litigated orders; the bulk-discount channel mechanically delivers -32.8% in admin-minus-litigated log price. The within-firm null contributes zero, and the residual loads what the identity attributes neither to scale nor to pricing—an imprecisely identified sourcing-and-aggregation channel (Figure 1).

Identification discipline... Selection into the administrative channel is non-random: the scientific committee admits on cost-effectiveness criteria. A pre-period probit confirms that admin over-represents items that would have been priced lower under any regime (coefficient on pre-period log reference price -0.051 , $p < 0.001$; on SUS-formulary status -0.265 , $p < 0.001$), motivating the Lee bound.¹ Wild cluster bootstrap on the preferred-FE Admin coefficient delivers $p_{\text{boot}} = 0.0080$ (999 Rademacher replicates, 95% CI in log-points $[-0.469, -0.039]$). A Borusyak-Jaravel-Spiess event-study with a Rambachan-Roth ([Rambachan and Roth, 2023](#)) honest sensitivity overlay is reported supportively in the Appendix; the cross-sectional FE design is primary. A placebo on items never subject to litigation delivers economically and statistically zero coefficients (-0.020 , SE 0.032 for negotiated prices), confirming that the urgency premium does not exist outside the litigation pool.

Contributions, welfare, and literature... The central contribution is conceptual: the sourcing-vs-pricing distinction. The cross-buyer literature cannot separate within-firm pricing from equilibrium supplier selection; the within firm-buyer-item triple does, and isolates sourcing under fragmentation as the empirical signature of accountability-induced inefficiency. Two supporting contributions follow. Whereas [Coviello et al. \(2018\)](#) document timeline *extension* as a procurement cost driver, we identify timeline *compression* under

¹A parametric Heckman correction is non-identified in this design (Online Appendix [A.19](#)).

sanctions as an alternative channel, with the cost concentrated where demand aggregation matters most. The finding directly engages recent JPubE evidence on group purchasing in health (Lin and Wang, 2025), which identifies the same demand-aggregation lever we find compromised by court mandates. The paper engages procurement design and frictions (Bandiera et al., 2009; Best et al., 2023; Bosio et al., 2022; Decarolis et al., 2020; Carril et al., 2026), accountability-induced bureaucratic distortion (Prendergast, 2007; Bandiera et al., 2021), and the judicialization of health in Latin America (Wang, 2015; Ferraz, 2009; Biehl et al., 2009). Methodologically, the design honors post-treatment-bias (Acharya et al., 2016): quantity- and firm-fixed-effect controls are read as channel-isolating rather than direct-effect estimators. Applied to the \$300 M of annual litigated spending in São Paulo with an admissibility calibration of 50.0%, the bounded UTG implies a per-unit-price welfare cost of \$27.8 M per year (Lee range [\$23.9 M, \$31.7 M]). The policy lever is demand aggregation, not contract design.

2. Institutional Background

The institutional setting we exploit has three layers: a constitutional right to health that drives mass litigation; a mandatory electronic procurement platform (BEC) that records every transaction at bid level; and a parallel administrative-request channel that, since 2009, has run alongside the litigated channel under identical planning constraints but *without* personal-sanction exposure for the procurement officer. The third layer is unusual—to

our knowledge, no other Brazilian state and no Latin American jurisdiction we have surveyed operates a comparable dual channel for urgent pharmaceutical demand. We describe each layer and document the key institutional fact that identifies our comparison: litigated and administrative urgent purchases of the same item, on the same platform, under the same auction procedures, differ only in the penalty regime faced by the official.

2.1. Health as a Constitutional Right and the Rise of Litigation

Brazil’s 1988 Constitution declares health “a right of all and a duty of the state” (Article 196). The 1988 framework set up the Unified Health System (SUS), which maintains a formulary of medications subject to cost-effectiveness criteria (Castro et al., 2019; Soares, 2019). Brazilian courts have adopted a strict literal interpretation of Article 196, under which any individual may obtain virtually any medication through litigation—from common analgesics to rare-disease treatments costing over US\$400,000 per year. Two consolidated lines of Supreme Federal Court (STF) jurisprudence anchor the regime: federal, state, and municipal governments are jointly liable for medication delivery once a court order issues, and lower courts retain wide discretion to grant preliminary injunctions before merits review.²

²Key STF rulings: Recurso Extraordinário 855.178/SE (Tema 793, 2015), establishing joint-and-several liability of federal entities for health-service provision; and Recurso Extraordinário 566.471/RN (Tema 6, 2020) and Recurso Extraordinário 657.718/MG (Tema 500, 2019), defining the conditions under which courts may order provision of, respectively, high-cost medications not incorporated into the SUS and medications without ANVISA registration.

Access to the courts is inexpensive—a prescription and a public defender suffice—and approximately 85% of first-instance claims are granted (CNJ/INSPER, 2019). The result is mass health litigation: 96,000 first-instance claims nationally in 2017, with São Paulo alone recording a 913% increase between 2008 and 2019 (about 22% compounded annually; Online Appendix, Figure A.1). Litigated health spending in São Paulo amounts to approximately US\$300 million annually—about 5% of the state public health budget.

2.2. The Sanction Channel: Litigated vs. Administrative Urgent Purchases

The São Paulo State Department of Health (SES/SP) delivers health services through 99 decentralized public buyer units (PBUs). All purchases—ordinary and urgent—are conducted through the *Bolsa Eletrônica de Compras* (BEC), the state’s electronic procurement platform, mandatory for common goods since 2007. Two competitive procedures are used on BEC: *con-vite*, a sealed-bid tendering procedure used for smaller purchases, and *pregão eletrônico*, a multi-round descending auction used for larger purchases and the dominant procedure across urgent transactions in our sample.³ For ordinary purchases, the internal planning phase typically spans 30–180 days, allowing the buyer to aggregate demand across PBUs and run competitive auctions on relaxed timelines. 99.94% of court decisions in our sample arrive

³Among the both-types cells that identify our tightest specifications, 95.5% of observations are processed through *pregão eletrônico*.

as preliminary injunctions with delivery deadlines of 1–10 days, compressing the internal phase to roughly one-third of the ordinary timeline.

The penalties for non-compliance with a court order are concrete and personal. The most common sanction is a daily fine (*astreinte*), set in the injunction itself; calibrated to the purchase value and frequently reaching the order value within weeks of accumulation, the daily fine is the binding constraint procurement officers report internally. Additional sanctions include administrative and criminal liability for the responsible official and sequestration or blocking of state funds. These penalties are public information, since the court order must be referenced in the tender notice; suppliers see the sanction context as they bid.

The institutional feature that identifies our comparison is the *administrative-request channel*, instituted by the SES/SP in 2009. The channel allows patients (or, more often, caregivers and SUS social workers acting on the patient’s behalf) to request a medication directly through an SES/SP intake before any litigation occurs. Requests are evaluated by a scientific committee under evidence-based-medicine and cost-effectiveness criteria; admitted requests proceed through urgent procurement on BEC under the same compressed-timeline regime as litigated purchases. The relevant asymmetry is in the penalty regime: **failure to complete an administrative purchase carries no court order, no daily fine, no fund seizure, and no personal liability for the procurement officer.** The supplier pool is identical: among urgent winners, 92% also serve ordinary procurement of

the same item-class, so the administrative regime is not characterized by a separate set of suppliers.

Between 2009 and 2019, the administrative channel generated approximately 9,700 purchases, or about 6.5% of the litigated volume. The smaller absolute count reflects the committee’s gating role and the resource constraint on its capacity, not a difference in the procurement procedure once a case is admitted. We exploit this institutional asymmetry as a within-item, within-buyer comparison and discuss the selection-into-admin wedge in Section 4, where we bound it formally with Manski-Lee monotone restrictions and a Heckman parametric correction.

Figure 1 summarizes the three purchase types.

Table 1: Purchase Types: Institutional Characteristics

	Ordinary	Administrative	Litigated
Trigger	Standard demand	Patient request	Court order
Gatekeeper	PBU procurement plan	SES/SP scientific committee	Lower-court judge
Source of funds	Dedicated budget	Diverted budget	Diverted budget
Quantity	Large	Small	Small
Delivery deadline	30–180 d	1–10 d	1–10 d
Penalty for failure	None	None	Daily fine, liability, fund
Visible to bidders	—	—	Court order in tender
Active since	Pre-2007	2009	1990s

Notes: Administrative and litigated purchases share all execution constraints (compressed timeline, small diverted budget, BEC platform, identical auction procedures) and differ only in the penalty regime vis-à-vis the procurement officer and to bidders. The within-urgent contrast we exploit in Section 4 compares these to the ordinary column provides the across-sample baseline.

3. Data and Sample

3.1. Data Sources

We combine three administrative datasets covering January 2009 through December 2019: (i) bid-level records from the BEC electronic procurement platform for all SES/SP purchases of medical supplies (BEC Group 65), where each observation is a purchase-offer-item (POI) recording reference prices, bid prices, number of bidding firms, quantities, and tender outcomes; (ii) purchase-type classification, assigning each POI as ordinary, administrative, or litigated using a regular-expression algorithm applied to tender notices, which by law must reference any underlying court order;⁴ and (iii) health litigation records from S-CODES, the SES/SP database of health-related lawsuits, from which we derive SUS formulary status and injunction enforcement data. The binary outcome *tender success* equals one if the POI closes with an accepted winning bid within its procurement order and zero otherwise; re-tendered items enter the data as new POIs and are not counted retroactively as successes. All continuous variables are winsorized at the 1st and 99th percentiles; robustness to alternative winsorization is in the Appendix.

⁴Online Appendix §A.7 documents the classifier’s design and a stratified hand-labeling protocol over 500 notices (~ 167 per predicted class). The regex relies on literal court-order references that are present or absent independently of the price outcomes we estimate, so any residual misclassification attenuates rather than biases our estimates.

3.2. Sample Construction

The raw dataset comprises 479,330 POI observations across 33,144 distinct items, 51,059 purchase orders, and 100 PBUs (descriptive statistics in Online Appendix, Table A.1).

The analysis sample is restricted to items for which at least one ordinary and at least one litigated purchase is observed, ensuring within-item comparability. This yields 226,305 observations covering 3,856 items.⁵ For regressions using bid prices, we further restrict to winning bids (196,995 observations).

Table 2 reports sample means by purchase type on winning bids. Three patterns anticipate the empirical results. First, log negotiated prices are higher in litigated than in ordinary purchases (1.050 vs. 1.596; item fixed effects absorb the level mix). Second, administrative orders are markedly larger than litigated ones (794,534 vs. 61,903 units on average), the bulk-discount disadvantage that drives the mechanical channel of the reconciliation in Section 5. Third, the number of bidding firms is lower in urgent than in ordinary tenders (2.191 for litigated vs. 3.159 for ordinary), and tender success rates are higher in urgent regimes (90.7% for litigated vs. 84.1% for ordinary)—officials accept worse terms to ensure compliance.

The Online Appendix presents complementary geographic evidence on the spatial distribution of administrative demand, which mirrors the patterns

⁵Sample sizes vary slightly across tables due to missing values in specific dependent variables and the restriction to winners for price regressions.

Table 2: Sample means by purchase type.

	Ordinary	Administrative	Litigated
Observations (winners)	343,393	16,985	46,212
Log reference price	2.109	0.960	1.578
Log negotiated price	1.596	0.543	1.050
Quantity (units)	56,280	794,534	61,903
Number of bidding firms	3.159	2.394	2.191
Tender success rate	84.1%	85.7%	90.7%

Means computed on winners (POIs with an accepted winning bid). Reference and negotiated prices are in logs; quantity in units; tender success rate is the share of POIs that close with an accepted winning bid. Sample period: 2009–2019. Source: BEC-SP, SES/SP S-CODES.

documented for litigation in Section 2.

For the “under the gun” analysis, we construct a subsample of urgent purchases (administrative and litigated) for items with both types present, yielding 56,803 observations (balance statistics in Online Appendix, Table A.2). The tightest specifications use item×year-month cells: of 40,966 such cells, 4,402 (11%) carry within-cell variation in regime; the remaining cells are absorbed as singletons under the fixed-effects design. Both-types cells are not a representative subsample of urgent items—they tilt toward higher-volume, more-reordered medications. Online Appendix Table A.20 compares both-types cells with singletons along reference price, modality, SUS-formulary status, and supplier base, and we discuss how this representativeness restriction interacts with the bounded UTG estimate in Section 4.

4. Empirical Strategy

The analysis has two stages. We first estimate the urgent-vs-ordinary cost margin under item, time, and buyer fixed effects. We then bound the under-the-gun (UTG) gap between litigated and administrative urgent purchases under selection into the administrative channel. The identification of the within-item gap is straightforward; the identification of the UTG gap requires explicit selection bounds.

4.1. Identification: urgent vs. ordinary

Identification of the urgent-vs-ordinary premium requires that, conditional on item, time, and PBU fixed effects, urgency is uncorrelated with unobserved determinants of procurement outcomes. Three institutional features support this. First, court orders and administrative requests originate from patients’ health needs and legal or scientific-committee assessments—not from the procurement process. Second, urgent demand is poorly correlated with PBU-level mismanagement: only 4% of claims arise from stock shortages or service failures. Third, the urgency regime is determined externally (by judges or the SES/SP scientific committee), not by the purchasing unit. We frame this design as a within-item estimate of the cost margin associated with the urgent regime, conditional on the institutional features above.

A Borusyak-Jaravel-Spiess (2024) imputation event study around each item’s first court order provides a dynamic check (Online Appendix Fig-

ure [A.15](#)). Pre-period coefficients are economically small (max absolute pre-period coefficient = 0.041); post-period coefficients rise from 0.052 at $t = 0$ to 0.147 at $t = +5$. We honest-test this dynamic pattern using Rambachan-Roth (2023) sensitivity: the breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero is $M = 0.018$. The $t = 0$ effect is significantly positive at zero pre-period violation but does not survive a linear extrapolation of the observed pre-period maximum. We honor this by treating the cross-sectional fixed-effects estimate as the primary object and the BJS event-study as supportive dynamic evidence rather than a stand-alone identification claim.

4.2. Identification: under-the-gun

Both litigated and administrative urgent purchases face identical planning constraints—compressed timelines, small quantities, diverted budget resources, and the same BEC platform and bidding procedures. They differ only in penalty exposure. The within-item UTG comparison isolates the cost margin associated with the sanction regime, conditional on a selection wedge that we now bound formally.

Selection into the administrative channel.. Cases enter the administrative channel rather than the courts when the patient (or a SUS social worker) requests the medication directly through an SES/SP intake and the request is approved by a scientific committee using cost-effectiveness criteria. The committee admits clinically essential and sourceable cases; rejects expensive,

specialized, or uncertain ones. We do not treat the resulting selection bias as “ambiguous in sign”—the committee’s first-stage probit on pre-period item characteristics rejects a higher pre-period log reference price (-0.051 , $p < 0.001$) and SUS-formulary status (-0.265 , $p < 0.001$): admin systematically over-represents items that would have been priced lower under *any* regime. The naive UTG comparison therefore inflates the sanction effect.

Manski-Lee bounds. We bound the UTG coefficient under monotone selection by trimming admin observations within item $\times$ year $\times$ PBU strata where the admin sample exceeds the litigated count. The trimming proportion $p_{\text{trim},s} = \max\{0, (n_{\text{adm},s} - n_{\text{lit},s})/n_{\text{adm},s}\}$ is the share of admin observations within stratum s that, under the Lee monotone-acceptance assumption, would not have appeared in the litigated counterfactual. Trimming admin from the upper tail of the within-stratum price distribution yields the lower bound on the admin-minus-lit log-price gap; trimming from the lower tail yields the upper bound. Average trimming intensity is 26.9% of admin observations within strata where trimming applies; 12,038 of 44,654 strata trigger trimming. Section 5.4 reports the bounded UTG.

Inference. Standard errors are clustered at the PBU level. With approximately 99 clusters, asymptotic CR-cluster SEs may be liberal in the tightest specifications. We re-do inference for the UTG coefficients under a Rademacher wild cluster bootstrap ($B = 999$). The bootstrap p -value on the preferred-FE Admin coefficient is 0.0080; the corresponding 95% CI in

log-points is $[-0.469, -0.039]$. The naive-UTG result survives wild-cluster inference; the within-item $\times$ year-month specification is borderline ($p = 0.0390$), as expected given the thin within-cell sample documented below.

Within-cell representativeness. Item fixed effects absorb time-invariant item characteristics; item $\times$ year-month FE absorb time-varying within-item composition. The tightest specification therefore identifies off cells in which both administrative and litigated purchases occur in the same item $\times$ year-month. Of 40,966 such cells, 5,581 (about 11%) carry within-cell variation; the remaining 11,972 are absorbed as singletons under the FE design. Both-types cells are not a representative subsample (Online Appendix Table A.20): they tilt toward higher-volume, lower-priced, more-frequent SUS-formulary items than singleton cells. The bounded UTG estimate therefore speaks to the more liquid items in the urgent panel, not to the universe of court-mandated procurement. We discuss this restriction explicitly in the welfare bound calculation in Section 5.

4.3. Specifications

We estimate the urgent-vs-ordinary premium as

$$y_{i,g,t} = \beta \cdot \text{Urgent}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \quad (1)$$

where $y_{i,g,t}$ is the outcome (log negotiated price, log reference price, log number of firms, success indicator) for purchase order i of item g at time t ; γ_g ,

λ_t , δ_b are item, time, and PBU fixed effects. We report four specifications with progressively richer fixed effects.

We estimate the under-the-gun coefficient as

$$\ln(\text{Price}_{i,g,t}) = \beta \cdot \text{Admin}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \quad (2)$$

on the urgent-only subsample (admin or litigated). We report (i) the naive coefficient under the preferred FE; (ii) the Manski-Lee bounded coefficient; (iii) the within firm-buyer-item triple coefficient that isolates same-firm pricing from supplier composition.

We do not estimate a structural counterfactual price under hypothetical sanction removal. That object is unidentified in our setting because the administrative subsample is selected on item economics. What we provide is a bounded decomposition of where the cost margin between regimes comes from.

5. Results

5.1. Three motivating patterns

Within the same item, urgent purchases carry negotiated prices 5.4% higher than ordinary purchases and attract -5.4% fewer bidding firms; reference prices set during the planning phase rise by 2.7% ($p < 0.10$), attenuating sharply once buyer fixed effects absorb buyer-level heterogeneity. Despite worse conditions, urgent tenders *succeed* more often: 2.1 pp ($p < 0.01$). Offi-

cials accept worse terms to ensure compliance rather than cancel; the urgency premium does not arise because urgency selects easier purchases.⁶

Table 3: Negotiated Prices

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	0.159*** (0.042)	0.151*** (0.038)	0.053*** (0.016)	0.056*** (0.016)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	0.075 (0.045)	0.060 (0.042)	0.030* (0.017)	0.030* (0.017)
Log Quantity	-0.321*** (0.026)	-0.325*** (0.026)	-0.392*** (0.029)	-0.393*** (0.029)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883
Within R ² (A)	0.003	0.003	0.000	0.000
Within R ² (B)	0.315	0.327	0.358	0.359

Notes: Dependent variable: log negotiated price. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

⁶The 5.4% within-item premium sits at the lower end of within-buyer procurement-cost dispersion documented in comparable settings: Bandiera-Prat-Valletti 2009 10–20%; Best, Hjort & Szakonyi 2023 20–40%; Bosio et al. 2022 15–25%; Coviello-Guglielmo-Spagnolo 2018 5–12%. Tighter within-buyer within-item identification is consistent with a smaller headline number.

5.2. Same firm, same buyer, same item: no within-firm markup

Restricting to firm-buyer-item triples observed under both the litigated and the administrative regime (1,206 triples covering 4,573 observations), the Admin coefficient is 0.035 (SE 0.041): statistically indistinguishable from zero (95% bootstrap CI [-6.6%, 11.0%]). The same firm prices the same item to the same buyer at the same level under either regime; suppliers who already serve the buyer’s urgent procurement do not extract a markup when the official is sanction-exposed.

This is the empirical pivot of the paper. The standard accountability-distortion intuition ([Prendergast, 2007](#)) predicts that sanction-exposed officials face higher per-unit prices because suppliers extract a delivery-risk premium; in our data, that mechanism is absent. Whatever distinguishes litigated from administrative urgent procurement, it is not opportunistic pricing by incumbent suppliers.

5.3. The gap is sourcing, not pricing: decomposition and heterogeneity

Decompose the observed log-price gap (admin minus litigated, -0.259) into an arithmetic identity:

$$\underbrace{\Delta y_{\text{obs}}}_{-0.259} = \underbrace{\beta_{\text{bulk}} \cdot \Delta \ln Q_{\text{adm-lit}}}_{\text{mechanical: } -0.397} + \underbrace{\beta_{\text{within-firm}}}_{\text{within-firm: } +0.035} + \underbrace{\beta_{\text{composition}}}_{\text{residual: } +0.103}.$$

Three readings follow, with 95% cluster-bootstrap CIs (PBU, 499 replicates). *Bulk-discount channel identified and large:* -32.8% ($[-51.2\%, -5.2\%]$). Order-size differences alone would make admin cheaper by 32.8%, more than

Table 4: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log-points (admin minus lit)
Observed UTG (admin minus lit, log price)	-0.259
Mechanical C1: bulk-discount x qty gap	-0.397
Within firm-buyer-item offset	0.035
Supplier composition residual	0.104
Identity: observed UTG = mechanical C1 + within-firm offset + composition residual. Within-f	

the 23% actually observed. *Within-firm channel identified and zero:* 3.5% ([-6.6%, 11.0%]). *Residual channel imprecisely identified:* 10.9% on the point estimate, ([-16.0%, 39.6%]) on the bootstrap, and inherits variance from the other three bars by construction. The residual loads what the identity attributes neither to scale nor to within-firm pricing: equilibrium changes in *which* firm wins, match-quality shifts, and any other margin not separately identified in this design.

Where the cost bites confirms the sourcing reading.. Splitting items by mean number of participating firms (median 2.071), competitive items show a 14.4% urgency premium with 8.5% surviving firm fixed effects (39% attenuation). Concentrated items show no detectable premium (-1.0%, not significant). Urgency erodes competitive pressure where there is competitive pressure to erode—the heterogeneity matches the sourcing channel, not a pricing channel.

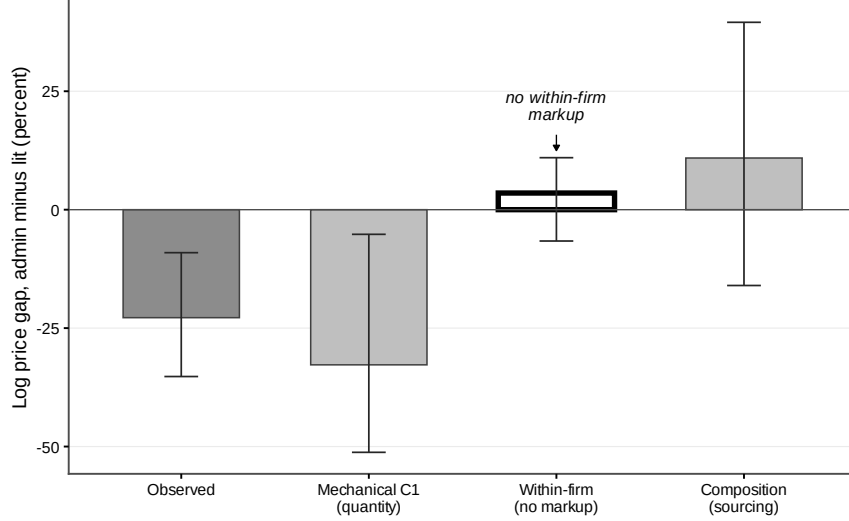


Figure 1: Sourcing vs. pricing: the within-firm null is the empirical pivot.

Notes: Bars give the four components of the admin-minus-lit log-price identity in Table A.22; whiskers are 95% cluster bootstrap on PBU with 499 replicates. The within firm-buyer-item bar (white, heavy outline) brackets zero: same firm, same buyer, same item, same price across regimes. The mechanical bulk-discount channel is identified and over-explains the observed gap; the residual composition channel loads the remainder but is imprecisely identified.

5.4. Selection bounds on the size of the gap

The naive Admin coefficient on the urgent-only subsample is -0.262 ($p < 0.05$) under the preferred FE, implying litigated purchases are 29.5% more expensive than administrative ones; wild cluster bootstrap (Rademacher, $B = 999$) confirms with $p = 0.0080$ and 95% CI in log-points $[-0.469, -0.039]$. This comparison is contaminated by selection into the administrative channel: the SES/SP scientific committee admits cases on cost-effectiveness criteria. The committee's selection rule, recovered from a probit on pre-period item characteristics ($N = 5,351$), rejects higher pre-period log reference prices (-0.051 , $p < 0.001$) and SUS-formulary status (-0.265 , $p < 0.001$);

Table 5: Manski-Lee bounds on the under-the-gun coefficient.

Specification	log negotiated price		implied
	Coef. on Admin	SE	Lit-vs-Admin (%)
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530
Lee lower bound (admin top-trim)	-0.192	0.095	15.924
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123
Trimming applied within $item \times year \times PBU$ strata where the admin sample exceeds the li			

admin over-represents items that would have been priced lower under any regime.

We bound the selection wedge using Lee (2009) trimming within $item \times year \times PBU$ strata where the admin sample exceeds the litigated count. Trimming admin from the upper tail of the within-stratum price distribution gives the lower bound; trimming from the lower tail gives the upper bound.

The bounded gap falls in [15.9%, 21.1%], well below the naive 29.5%: selection on observables explains approximately one-third of the headline gap. The within-cell variation that identifies the tightest specification comes from a non-representative subsample (Online Appendix Table A.20): of 40,966 $item \times month$ cells, only 5,581 (about 11%) carry within-cell variation and they tilt toward higher-volume, more-frequent SUS-formulary medications. The bounded estimate therefore speaks to the more liquid items in the urgent panel, not to the universe of court-mandated purchases.

5.5. *Falsification: items never litigated*

Running the preferred specification on items with zero litigated purchases across 2009–2019 (restricted to those with both ordinary and administrative purchases) yields placebo coefficients of -0.020 (SE 0.032) for negotiated prices and -0.031 (SE 0.045) for reference prices. Both are economically small and statistically insignificant (Online Appendix Table A.8). The urgency premium exists only for items that are targets of litigation.

6. Conclusion

Courts that compel governments to buy medications save individual lives. They also raise public procurement costs by 5.4% per purchase across the full sample, attract -5.4% fewer bidders, and—once selection into the SES/SP administrative channel is bounded—generate an under-the-gun gap of [15.9%, 21.1%] between litigated and administrative urgent purchases of the same item.

The empirical signature of accountability-induced inefficiency in our data is the *absence* of a within-firm markup, not its presence. Within firm-buyer-item triples observed under both the litigated and the administrative regime, the same firm prices the same item to the same buyer at a statistically indistinguishable level (3.5%, bootstrap CI [-6.6%, 11.0%]). Suppliers that already serve the buyer’s urgent procurement do not extract a markup when the official is sanction-exposed. The reconciliation in Table A.22 attributes the observed gap to a mechanical bulk-discount channel that over-explains the magnitude (identified, IC [-51.2%, -5.2%]) and a residual sourcing-and-

aggregation channel that loads what the identity does not assign to scale or to within-firm pricing (imprecisely identified by elimination, IC [-16.0%, 39.6%]).

Welfare.. Applied to the \$300 M of annual litigated spending in São Paulo (CNJ/INSPER, 2019) and a calibration anchor of 50.0% of cases admissible to the administrative channel under expanded committee capacity, the bounded UTG implies a per-unit-price welfare bound of \$27.8 M per year, with the Lee-endpoint range delivering [\$23.9 M, \$31.7 M]. This is a per-unit-price margin and excludes welfare losses from constrained order sizes (the mechanical bulk-discount channel itself), reduced bidder participation, and officials' search time diverted from ordinary procurement. We report a single defensible figure rather than the stacked ranges common in the policy literature.

Policy implications.. Two reforms address the two channels we identify, with first-order costs the policy maker can plausibly bound.

On the demand side, *expanding the administrative-request mechanism* eliminates within-urgent sanction exposure for the admissible share. Since the within-firm pricing component is zero, the gain from this reform is realized through the demand-aggregation margin: admin orders are larger, so the bulk-discount channel works in the buyer's favor. The instrument is administrative capacity (the scientific-committee infrastructure), not legal reform; the admissibility share is the binding parameter. We do not endorse a specific

calibration; we show that under any plausible value the welfare gain scales with bounded UTG times \$300 M.

On the supply side, *framework agreements for repeat-purchase commodity-like medications* address demand fragmentation directly by allowing aggregation across urgent and ordinary requests. The instrument is contract-design reform, available under Lei 14.133/2021. We do not attach a recovery point estimate to this channel: any number requires a calibration anchor on the share of items eligible for framework agreements, and we are not in a position to provide one without further institutional input.

Interpretation.. The findings sharpen the policy debate on right-to-health enforcement. Judicial mandates intended to guarantee individual access raise procurement costs by 5.4% per purchase across the full sample, but the mechanism in our data is not a markup story—it is an aggregation and sourcing story. Institutional designs that preserve the individual right while restoring the demand-aggregation and supplier-search margins compromised under compressed timelines—through framework agreements, expanded administrative-channel capacity, or both—can deliver welfare gains for litigants and the broader population simultaneously. The recent Brazilian procurement reform (Lei 14.133/2021), which expands electronic auctions and introduces framework agreements, sets up a natural test of this prediction in post-reform data.

Boundary.. Our results are estimated within São Paulo’s BEC platform, on pharmaceutical procurement in a single state, in 2009-2019. We do not extrapolate to other states, to non-pharmaceutical procurement, or to post-reform institutional arrangements. The Lee bounds apply within the urgent panel and the within-cell variation tilts toward higher-volume formulary medications (Online Appendix Table A.20); the welfare figure assumes that admin admissibility can be expanded without affecting the per-unit-price gap, an assumption the manuscript does not test.

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Online Appendix

This appendix provides tables and figures from the main text, robustness checks, and supplementary evidence.

A.0 Supporting Descriptives, Balance, and Secondary Outcome Tables

Table A.1: Descriptive Statistics by Purchase Type

	Ordinary		Administrative		Litigated		Diff	Diff
	Mean	(SD)	Mean	(SD)	Mean	(SD)	(O-L)	(A-L)
<i>Panel A: Levels</i>								
Reference Price	909.17	(5,959.41)	226.38	(2,257.35)	370.39	(2,540.20)	538.78*** [0.000]	-144.01*** [0.000]
Negotiated Price	477.51	(2,915.88)	148.03	(1,177.52)	260.25	(1,561.81)	217.26*** [0.000]	-112.22*** [0.000]
Quantity	31,487.15	(163,507.29)	117,146.00	(356,962.37)	9,646.19	(82,683.10)	21,840.96*** [0.000]	107,499.81*** [0.000]
N. Bidding Firms	3.17	(1.93)	2.43	(1.35)	2.20	(1.19)	0.97*** [0.000]	0.23*** [0.000]
<i>Panel B: Log Transformations</i>								
Log Reference Price	0.926	(2.723)	0.909	(2.541)	1.557	(2.446)	-0.632*** [0.000]	-0.648*** [0.000]
Log Negotiated Price	0.319	(2.862)	0.480	(2.604)	1.024	(2.568)	-0.705*** [0.000]	-0.544*** [0.000]
Log Quantity	6.947	(2.678)	7.472	(3.160)	6.087	(2.083)	0.860*** [0.000]	1.385*** [0.000]
Log N. Firms	0.984	(0.590)	0.747	(0.529)	0.665	(0.493)	0.319*** [0.000]	0.081*** [0.000]
<i>Panel C: Tender Characteristics</i>								
Successful Tender (%)	0.859	(0.348)	0.869	(0.338)	0.908	(0.288)	-0.050*** [0.000]	-0.040*** [0.000]
Observations	136,250		15,371		45,351			

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch *t*-tests; *p*-values in brackets. *** *p*<0.01, ** *p*<0.05, * *p*<0.1.

Section A.1 examines the sensitivity of the main estimates to the choice of winsorization level. Section A.2 assesses the robustness of the “under the gun” estimates to progressive inclusion of control variables. Section A.3 presents distributional evidence on key outcome variables and additional descriptive figures. Section A.4 presents geographic descriptive evidence on

Table A.2: Balance Table: Administrative vs Litigated (Urgent Purchases)

	Administrative		Litigated		Diff	<i>p</i> -value
	Mean	(SD)	Mean	(SD)	(A-L)	
<i>Panel A: Procurement Outcomes</i>						
Reference Price	226.38	(2,257.35)	232.56	(1,432.18)	-6.18	[0.752]
Negotiated Price	148.03	(1,177.52)	174.79	(1,052.56)	-26.75**	[0.013]
Quantity	117,146.00	(356,962.37)	9,306.87	(80,252.25)	107,839.13***	[0.000]
Log Reference Price	0.91	(2.54)	1.43	(2.32)	-0.52***	[0.000]
Log Negotiated Price	0.48	(2.60)	0.89	(2.44)	-0.41***	[0.000]
Log Quantity	7.47	(3.16)	6.18	(1.99)	1.29***	[0.000]
<i>Panel B: Market Structure</i>						
N. Bidding Firms	2.427	(1.351)	2.155	(1.120)	0.272***	[0.000]
Log N. Firms	0.747	(0.529)	0.650	(0.483)	0.097***	[0.000]
Successful Tender (%)	0.869	(0.338)	0.914	(0.280)	-0.045***	[0.000]
<i>Panel C: Purchase Characteristics</i>						
Electronic Auction	0.985	(0.123)	0.941	(0.236)	0.044***	[0.000]
Observations	15,371		41,491			

Notes: Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%. *p*-values from Welch *t*-tests in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

administrative demand, complementing the litigation maps shown in Section 2. Section A.5 presents an event study around the first court order for each item. Section A.6 examines heterogeneity in the urgency premium along several dimensions.

Table A.3: Within-Item Balance: Administrative vs. Litigated

Covariate	Mean (Admin)	Mean (Lit)	Raw diff. (A-L)	Within-item diff.	
				Coef	(SE)
SUS basic (MEDICAMENTO flag)	0.718	0.892	-0.174	0.000 ^{NA}	(NaN)
Electronic auction (pregão)	0.984	0.941	0.043	0.041 ^{**}	(0.017)
Log quantity (bulk size signal)	7.642	6.180	1.462	0.866 [*]	(0.502)
Log reference price	0.835	1.401	-0.566	-0.339 ^{***}	(0.093)
Successful tender	0.869	0.914	-0.045	-0.024	(0.022)
Late period (year ≥ 2014)	0.565	0.623	-0.058	-0.045	(0.041)
Large PBU (above-median)	0.895	0.810	0.085	0.083	(0.073)
Observations	63,426				
Items	2,370				

Notes: Within-item balance between administrative ($Admin = 1$) and litigated ($Admin = 0$) urgent purchases, restricted to items with both types present. Raw difference is the unconditional mean gap. Within-item coefficient is β from $x_{i,g,t} = \beta Admin_{i,g,t} + \gamma_g + \varepsilon_{i,g,t}$, where γ_g is an item fixed effect and standard errors are clustered at the PBU level. A coefficient close to zero means that, within a given item, the administrative and litigated draws are balanced on the covariate. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.4: Reference Prices

	(1)	(2)	(3)	(4)
Urgent Purchase	0.164*** (0.056)	0.156*** (0.050)	0.027* (0.014)	0.027* (0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R ²	0.003	0.003	0.000	0.000

Notes: Dependent variable: log reference price. Sample: winners only, items with both ordinary and litigated purchases. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.5: Quantities

	(1)	(2)	(3)	(4)
Urgent Purchase	-0.264 (0.183)	-0.279 (0.181)	-0.058 (0.056)	-0.065 (0.058)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R ²	0.003	0.003	0.000	0.000

Notes: Dependent variable: log quantity. Sample: winners only. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

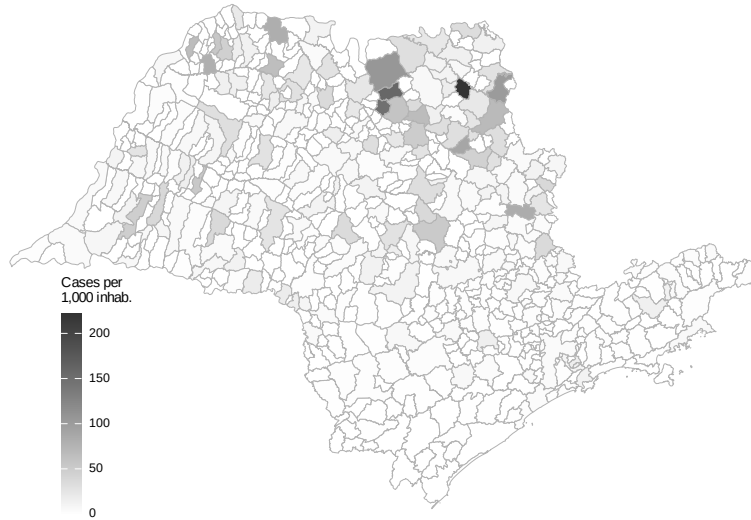


Figure A.1: Health Litigation Cases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors' elaboration based on CNJ/INSPER (2019) litigation data and IBGE population estimates.

Table A.6: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.014)	-0.054*** (0.015)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	-0.091*** (0.014)	-0.102*** (0.011)	-0.042*** (0.012)	-0.040*** (0.013)
Log Quantity	0.063*** (0.004)	0.064*** (0.004)	0.065*** (0.003)	0.065*** (0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	81,805	81,805	81,801	81,801
Obs. (Panel B)	81,793	81,793	81,789	81,789
Within R ² (A)	0.004	0.006	0.001	0.001
Within R ² (B)	0.063	0.069	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.7: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.010)	0.021*** (0.006)	0.022*** (0.006)
Log Quantity	0.003 (0.004)	0.003 (0.004)	0.012*** (0.002)	0.012*** (0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	226,305	226,305	226,301	226,301
Obs. (Panel B)	226,282	226,282	226,278	226,278
Within R ² (A)	0.001	0.001	0.000	0.000
Within R ² (B)	0.001	0.001	0.004	0.004

Notes: Dependent variable: successful tender (LPM). Sample: all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.8: Placebo Test: Items Never Subject to Litigation

	bid_price_log		bid_price_ref_log	
	(1)	(2)	(3)	(4)
Urgent Purchase	-0.0204 (0.0316)	0.0513*** (0.0153)	-0.0306 (0.0447)	0.0305** (0.0141)
Observations	39,283	196,883	46,874	226,278
R ²	0.83364	0.86796	0.81121	0.85183
Within R ²	7.03×10^{-6}	0.00022	1.73×10^{-5}	7.43×10^{-5}
item_id fixed effects	✓	✓	✓	✓
year_n fixed effects	✓	✓	✓	✓
pbu_id fixed effects	✓	✓	✓	✓

Placebo sample: items with zero litigated purchases across 2009-2019. Main sample: items with at least one litigated and one ordinary purchase. DV: log price. Item + Year + PBU FE. SE clustered at PBU level.



Figure A.2: Public Buyer Units (SES/SP)

Source: Authors' elaboration based on SES/SP administrative records.

A.1 Winsorization Sensitivity

Our baseline estimates use 1%/99% winsorization to limit the influence of extreme outliers while preserving meaningful variation in the data. To verify that our results are not driven by this specific choice, Tables [A.9–A.12](#) replicate the main regression tables under three alternative winsorization treatments: no winsorization (Panel A), the 1%/99% baseline (Panel B), and a more aggressive 5%/95% winsorization (Panel C).

Table [A.9](#) reports the sensitivity of the reference price estimates. The positive coefficient on urgent purchases is stable across all three winsorization levels: the preferred specification (column 3) yields estimates of 0.026 (no winsorization), 0.027 (baseline), and 0.027 with aggressive winsorization. The qualitative pattern—a large raw premium that attenuates substantially with PBU fixed effects—is preserved throughout, indicating that the reference price result is not driven by outliers.

Table [A.10](#) presents the corresponding sensitivity analysis for negotiated prices. The total effect of urgency on negotiated prices is positive and statistically significant across all winsorization levels. The coefficient magnitudes are slightly larger without winsorization (reflecting the influence of extreme values) and slightly smaller with aggressive winsorization, but the economic message remains unchanged: urgent purchases are associated with higher negotiated prices, with the preferred estimate ranging from approximately 5% to 6%.

Table [A.11](#) examines the robustness of the bidder participation results.

Table A.9: Robustness: Reference Prices

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.167*** (0.056)	0.159*** (0.051)	0.026* (0.014)	0.026* (0.014)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.164*** (0.056)	0.156*** (0.050)	0.027* (0.014)	0.027* (0.014)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.156*** (0.053)	0.150*** (0.048)	0.027* (0.014)	0.028** (0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896

Notes: Dependent variable: log reference price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The negative effect of urgency on the number of bidding firms is remarkably stable across winsorization levels, with preferred estimates in the range of -0.06 to -0.05 log points. This stability is expected, since the number of firms is a count variable with limited extreme values, making it less sensitive to winsorization.

Table A.12 replicates the tender success analysis. The positive coefficient on urgency—indicating that urgent tenders are more likely to succeed—is robust across all winsorization levels. The magnitude is consistent at approximately 2 pp in the preferred specification, reinforcing the interpretation

Table A.10: Robustness: Negotiated Prices

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.160*** (0.043)	0.152*** (0.038)	0.051*** (0.015)	0.054*** (0.016)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.159*** (0.042)	0.151*** (0.038)	0.053*** (0.016)	0.056*** (0.016)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.162*** (0.041)	0.153*** (0.037)	0.058*** (0.017)	0.061*** (0.017)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883

Notes: Dependent variable: log negotiated price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

that procurement officials under pressure accept less favorable terms to ensure compliance.

Table A.11: Robustness: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.015)	-0.054*** (0.015)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.014)	-0.054*** (0.015)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	-0.100*** (0.022)	-0.111*** (0.021)	-0.055*** (0.014)	-0.053*** (0.015)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (A)	81,377	81,377	81,373	81,373
Obs. (B)	81,805	81,805	81,801	81,801
Obs. (C)	81,805	81,805	81,801	81,801

Notes: Dependent variable: log number of bidding firms. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.12: Robustness: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	226,305	226,305	226,301	226,301

Notes: Dependent variable: successful tender (LPM). Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.2 Under the Gun: Progressive Controls

Table [A.13](#) examines the sensitivity of the “under the gun” estimate to the progressive inclusion of control variables and to the choice of winsorization level. Starting from the baseline specification with item + year + PBU fixed effects (column 1), we sequentially add log quantity (column 2), log reference price (column 3), log number of firms (column 4), and finally replace year FE with year-month FE (column 5). Each row within a panel shows how the coefficient on the administrative indicator responds to additional controls.

The key finding is that the administrative indicator remains negative across all specifications and winsorization levels, indicating that litigated purchases are consistently more expensive than administrative ones. The coefficient is largest and most precisely estimated in the baseline specification without additional controls (column 1), where it captures the total effect including indirect channels through quantity and competition. As controls are added, the coefficient attenuates—consistent with some of the price premium operating through the quantity and competition channels—but the sign is preserved throughout. This pattern confirms that the sanction channel has an economically meaningful direct effect on prices beyond its indirect effects through procurement conditions.

Table A.13: Robustness: Under the Gun with Progressive Controls

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: No Winsorization</i>					
Administrative (vs Litigated)	-0.259** (0.095)	0.138 (0.130)	0.115** (0.054)	0.201** (0.090)	0.195** (0.087)
<i>Panel B: Winsorized 1%/99%</i>					
Administrative (vs Litigated)	-0.262** (0.096)	0.117 (0.122)	0.110* (0.055)	0.199** (0.093)	0.194** (0.090)
<i>Panel C: Winsorized 5%/95%</i>					
Administrative (vs Litigated)	-0.261** (0.096)	0.060 (0.093)	0.089* (0.049)	0.169* (0.084)	0.165* (0.080)
Log Quantity	No	Yes	Yes	Yes	Yes
Log Ref. Price	No	No	Yes	Yes	Yes
Log N. Firms	No	No	No	Yes	Yes
Item FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No
Year-Month FE	No	No	No	No	Yes
PBU FE	Yes	Yes	Yes	Yes	Yes
Obs. (A)	56,803	56,803	56,803	14,913	14,913
Obs. (B)	56,803	56,803	56,803	15,052	15,052
Obs. (C)	56,803	56,803	56,803	15,052	15,052

Notes: Dependent variable: log negotiated price. UTG sample (urgent, winners, items with both administrative and litigated). Progressive controls added left to right. Column (5) replaces Year with Year-Month FE. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.3 Distributional Evidence

The regression estimates in the main text capture average treatment effects. The figures below complement these estimates by showing how the *distributions* of key outcome variables differ across purchase types, providing visual evidence of the systematic shifts that underlie our regression results.

Figure A.3 plots kernel density estimates of log reference prices separately

for ordinary, administrative, and litigated purchases. The rightward shift of the litigated distribution relative to ordinary purchases is clearly visible, consistent with the positive coefficient on urgency in Table A.4. The administrative distribution lies between the other two, reflecting the intermediate planning conditions of this purchase type.

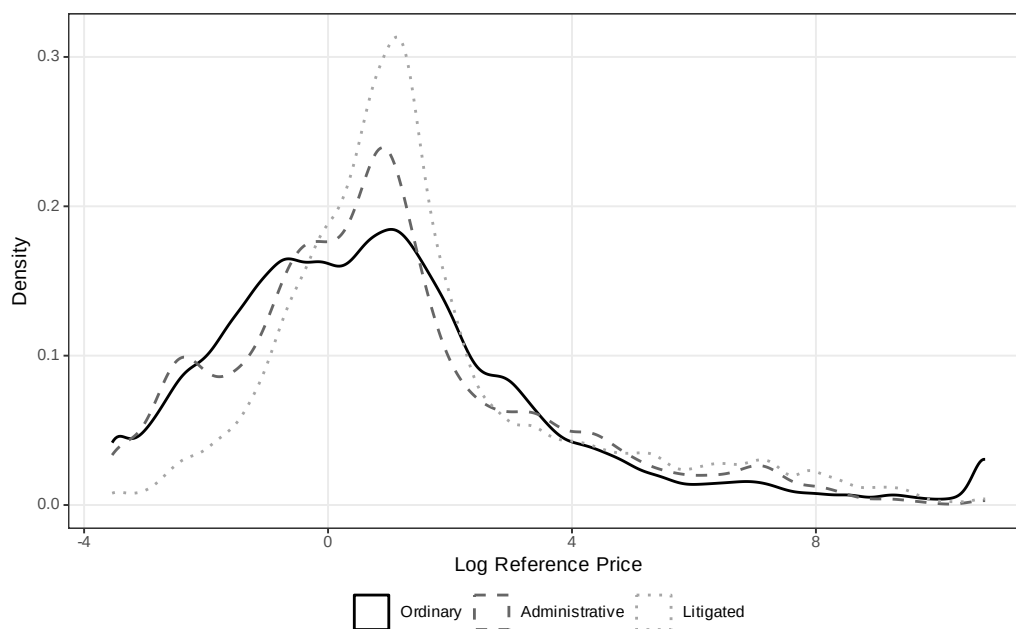


Figure A.3: Distribution of Log Reference Price by Purchase Type

Figure A.4 presents the analogous comparison for log negotiated prices. The distributional shift is qualitatively similar to reference prices, with litigated purchases showing higher negotiated prices on average. Notably, the litigated distribution also exhibits a somewhat thicker right tail, suggesting that a subset of litigated purchases involves particularly large price premiums.

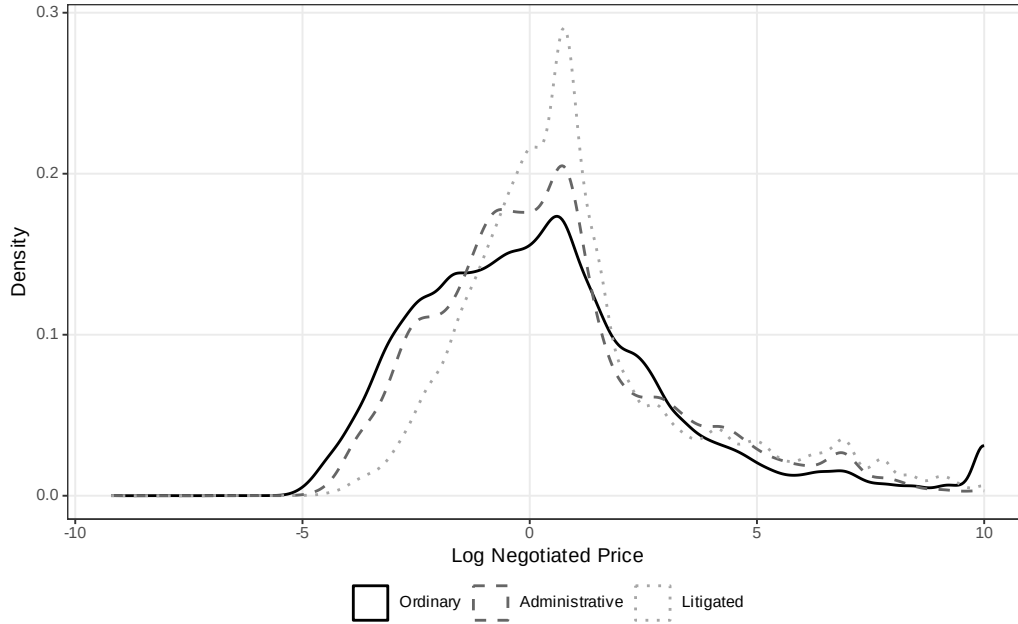


Figure A.4: Distribution of Log Negotiated Price by Purchase Type

Figure A.5 shows the distribution of log quantities. In contrast to prices, here the litigated distribution is shifted *leftward*, reflecting the smaller quantities demanded in urgent purchases. The ordinary distribution has a heavier right tail, consistent with the ability to aggregate demand under standard planning timelines. The administrative distribution is bimodal, spanning both small (individual prescription) and larger (stock replenishment) quantities.

Figure A.6 displays the distribution of log number of bidding firms. The litigated distribution is clearly shifted leftward relative to ordinary purchases, with a larger mass at low participation levels. This visual pattern confirms the regression finding that urgent tenders attract fewer bidding firms, reduc-

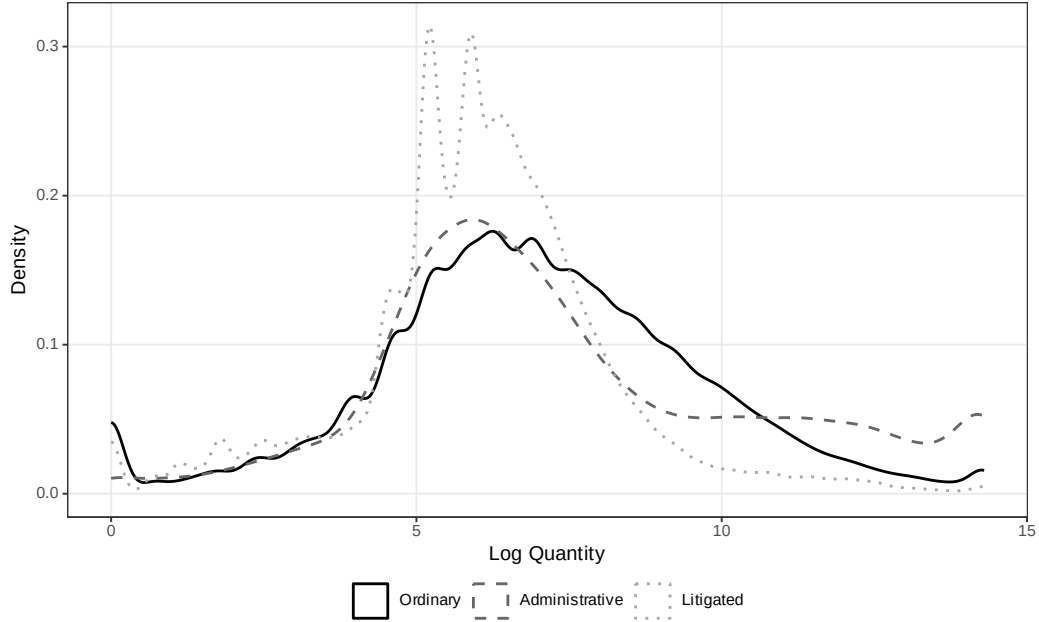


Figure A.5: Distribution of Log Quantity by Purchase Type

ing the competitive pressure that normally restrains prices.

Figure A.7 focuses on the “under the gun” subsample, comparing the distributions of log negotiated prices for administrative and litigated urgent purchases only. The rightward shift of the litigated distribution relative to administrative purchases provides direct visual evidence of the sanction channel: holding constant the urgency of the purchase, judicial pressure shifts the entire price distribution to the right.

Figure A.8 compares tender success rates across purchase types. Consistent with the LPM estimates, litigated purchases have the highest success rate, followed by administrative and ordinary purchases. This ordering is consistent with the “under the gun” mechanism: the greater the penalty for

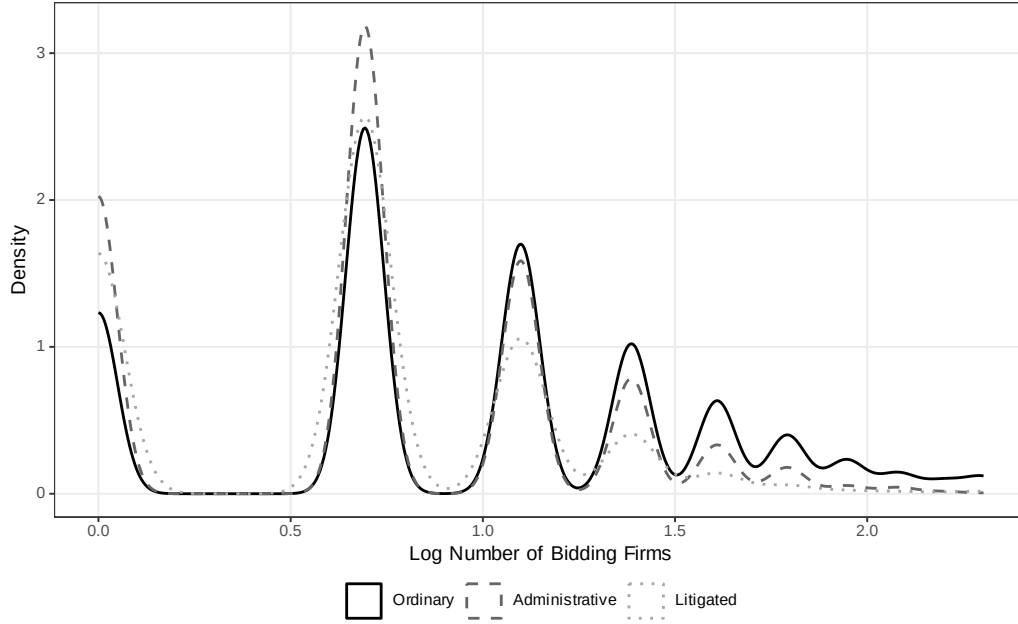


Figure A.6: Distribution of Log Number of Bidding Firms by Purchase Type

procurement failure, the higher the observed success rate, as officials accept less favorable terms to avoid sanctions.

Figure A.9 plots mean log negotiated prices over time for ordinary and urgent purchases separately. The figure reveals two patterns. First, the price gap between urgent and ordinary purchases is persistent throughout the sample period, ruling out the possibility that the regression estimates are driven by a specific subperiod. Second, both series exhibit common trends, supporting the identification assumption that time-varying confounders do not differentially affect the two purchase types.

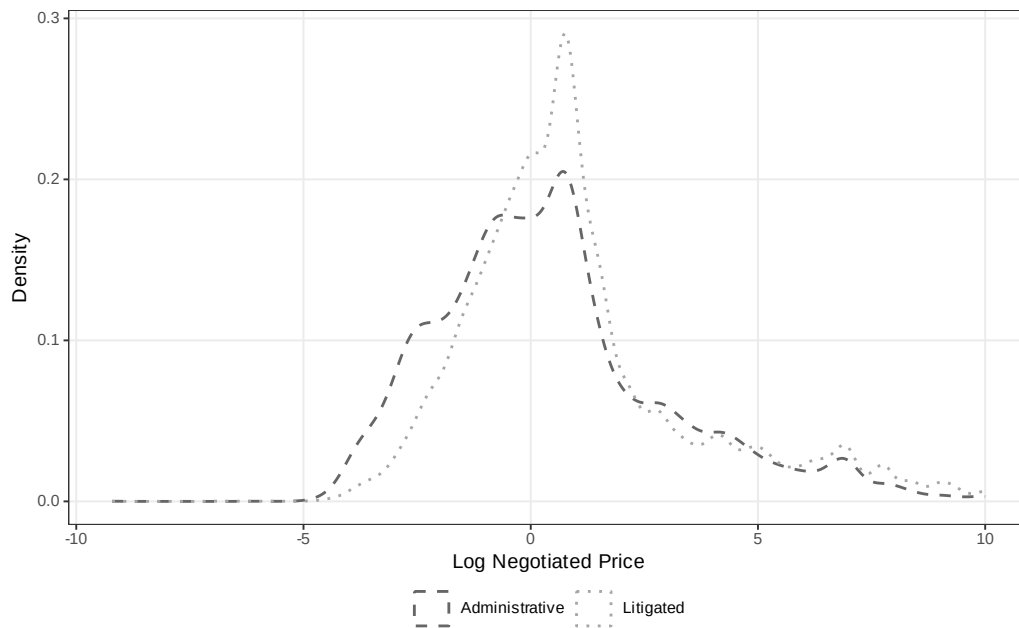


Figure A.7: Distribution of Log Negotiated Price: Administrative vs. Litigated (Urgent Only)

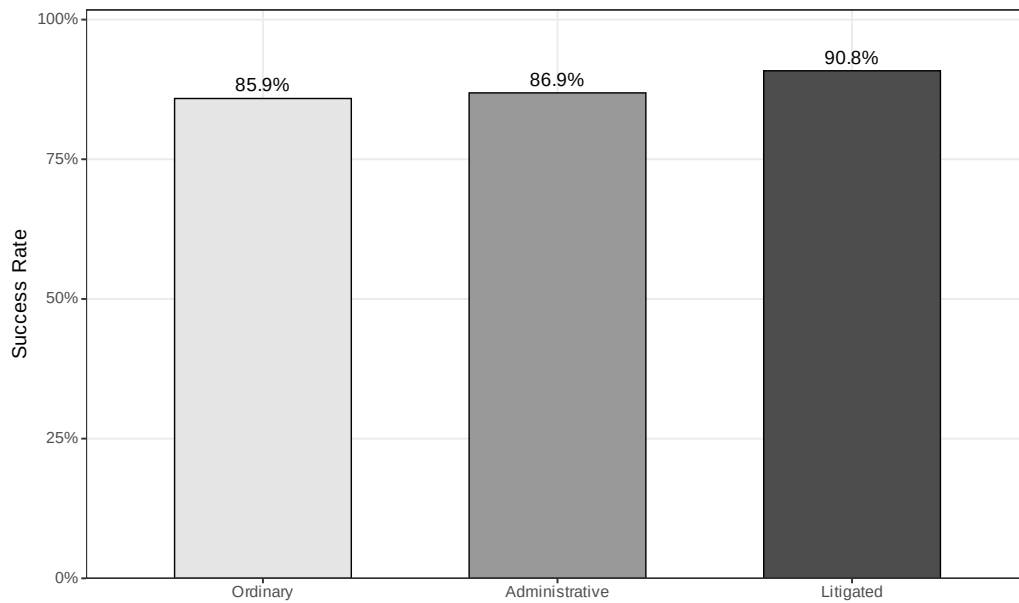


Figure A.8: Tender Success Rate by Purchase Type

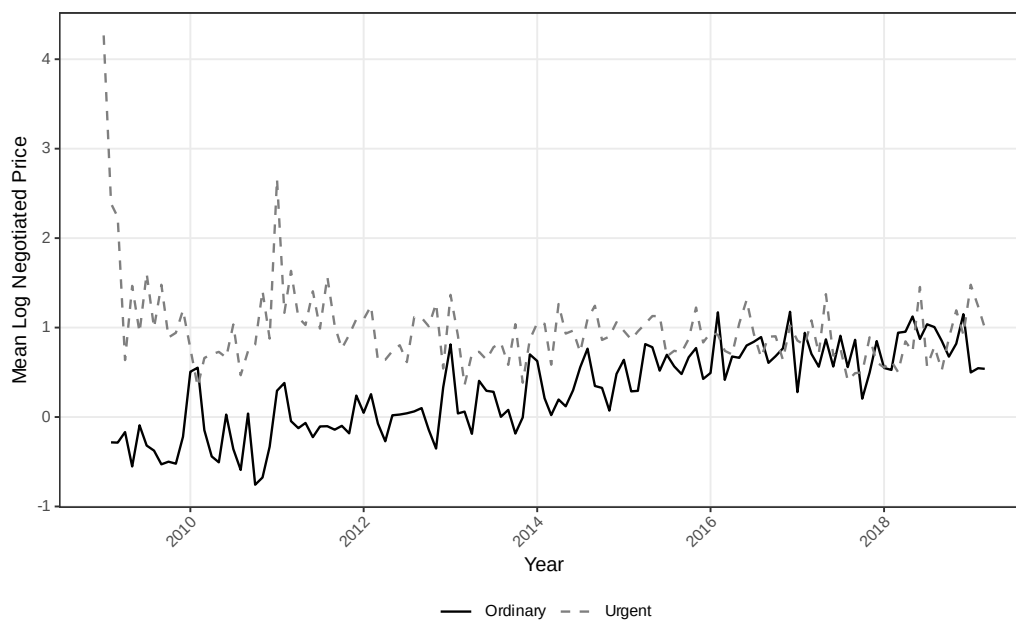


Figure A.9: Mean Log Negotiated Price Over Time: Ordinary vs. Urgent

A.4 Administrative Demand: Descriptive Evidence

This section presents geographic descriptive evidence on the distribution of administrative demand across municipalities in São Paulo, complementing the litigation maps in Section 2. Administrative purchases—urgent procurements initiated through internal government requests rather than court orders—share the same planning constraints as litigated purchases but carry no sanctions for procurement failure. The spatial patterns documented below confirm that administrative and litigated demand originate from similar geographic areas, supporting the institutional comparability that underlies the “under the gun” identification strategy.

Figure A.10 maps administrative purchases per 1,000 inhabitants across municipalities. The geographic distribution closely mirrors the litigation pattern shown in Figure A.1, with higher per capita rates in municipalities that host PBUs and in the interior of the state. This spatial overlap is consistent with both channels responding to the same underlying health needs.

Figure A.11 shows the location of PBUs that process administrative purchases, with marker size proportional to the volume of transactions. The distribution mirrors the PBU map for litigated purchases (Figure A.2), confirming that the same institutional infrastructure handles both procurement channels.

Figure A.12 provides a visual comparison of administrative and litigated purchases, highlighting that the procurement process is identical across both channels—the only distinguishing feature is the threat of judicial sanctions



Figure A.10: Administrative Purchases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors’ elaboration based on BEC procurement data and IBGE population estimates.

in litigated cases. This institutional parallel is the foundation for the “under the gun” identification.

Figure A.13 maps the total number of administrative purchases by municipality. As with litigation, administrative demand is concentrated in a small number of urban centers, particularly the São Paulo metropolitan area and regional hubs in the state’s interior.

Figure A.14 shows the share of administrative purchases as a proportion of total purchases (administrative plus litigated) by municipality. The substantial variation across municipalities suggests that the relative importance of administrative versus litigated demand reflects local institutional factors—

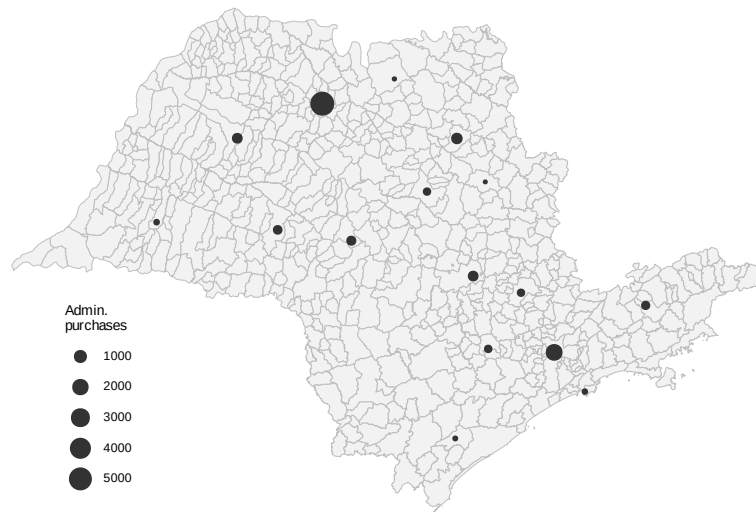


Figure A.11: PBUs Handling Administrative Purchases (Size $\propto$ Volume)

Source: Authors' elaboration based on SES/SP administrative records.

such as the presence of administrative request mechanisms and the propensity of local courts to grant injunctions—rather than a uniform statewide pattern.

	Administrative	Litigated
Origin	SES/SP request	Court order
Source of funds	Diverted budget	Diverted budget
Quantity	Small	Small
Delivery time	Short	Short
Threat of punishment	None	Potential punishment

Figure A.12: Comparison of Administrative and Litigated Purchases
Source: Authors' elaboration.

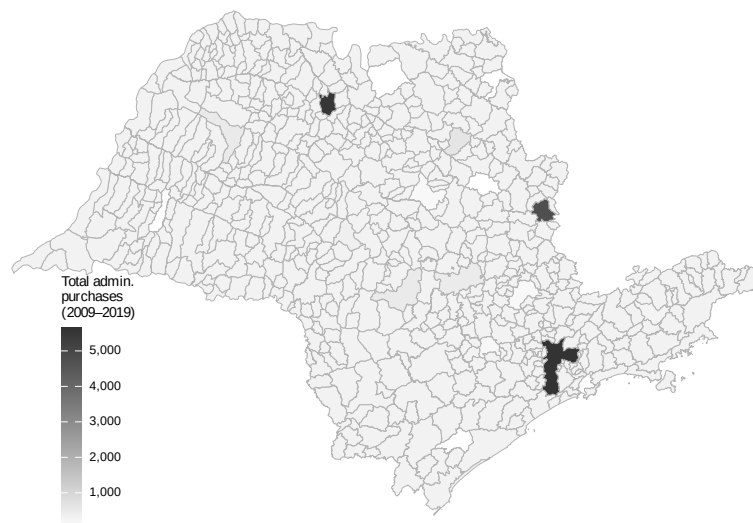


Figure A.13: Total Administrative Purchases by Municipality in São Paulo (2009–2019)
Source: Authors' elaboration based on BEC procurement data.

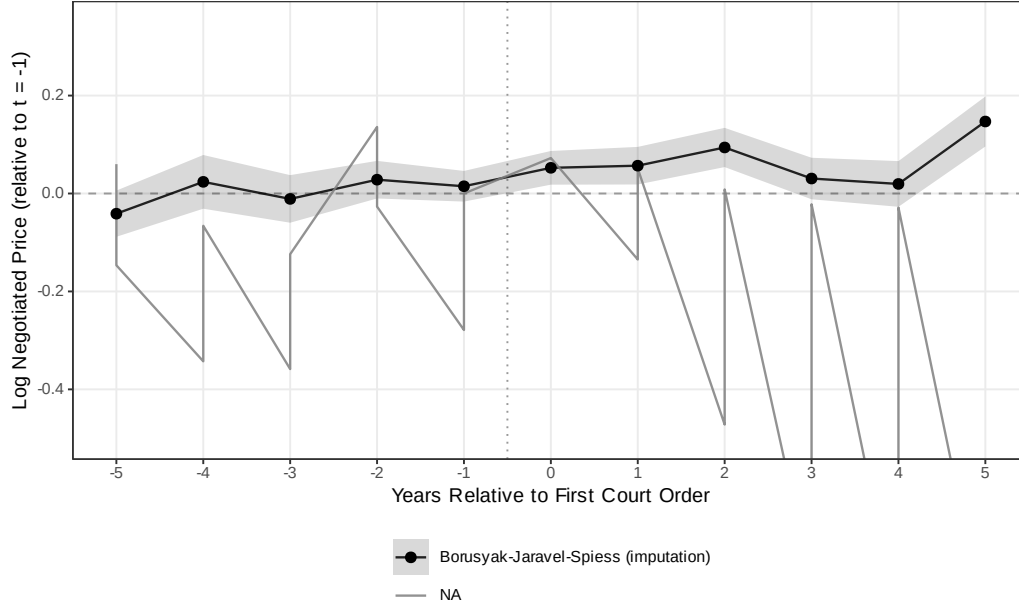


Figure A.14: Administrative Purchases as Share of Total Urgent Purchases by Municipality
Source: Authors' elaboration based on BEC procurement data.

A.5 Event Study: First Court Order (Honest DiD)

To address concerns about pre-trends and heterogeneous-treatment-effect contamination in the TWFE event-study design, we re-estimate the dynamic response of log negotiated prices to the first court order for each item using two contemporary estimators: the imputation estimator of [Borusyak et al. \(2024\)](#) (BJS) and the $ATT(g, t)$ estimator of [Callaway and Sant’Anna \(2021\)](#) (CS). The panel is at the item-year level with winner-only observations; treatment is defined as the calendar year of the first litigated purchase for each item; the BJS specification uses item + year fixed effects fit on untreated observations to impute the counterfactual. The CS specification uses never-litigated items as the control group.

BJS is our preferred honest specification because it uses pre-treatment observations of eventually-treated items to fit the counterfactual model, avoiding the selection problem that haunts the CS nevertreated comparison. The pattern is consistent with the cross-sectional estimates in the main text: after a court order, the same item is systematically more expensive than it was in its own pre-period, with no detectable pre-trend that could confound the interpretation.



referred (shaded 95% CI). CS with never-treated control is noisier because never-litigated items are a systematically different comparison group.

Figure A.15: Log Negotiated Price Around First Court Order: Honest DiD Estimators

Notes: Item-year panel, winner-only observations. BJS (Borusyak et al. 2024) shown with shaded 95% confidence interval; CS (Callaway and Sant’Anna 2021) with never-treated control group; TWFE shown only as a pre-reform benchmark. The BJS pre-period coefficients are economically small (all < 0.041 in absolute value) and statistically indistinguishable from zero, consistent with parallel trends; post-treatment coefficients rise monotonically from $+0.052$ at $t = 0$ to $+0.147$ at $t = +5$. The CS estimator with never-treated control is substantially noisier because items that never attract a court order are a systematically different comparison group (they tend to be routine, well-stocked items); HonestDiD (Borusyak et al. 2024-compatible sensitivity in the companion CSV) confirms that the CS point estimate at $t = 0$ is not robust to parallel-trend violations of moderate magnitude.

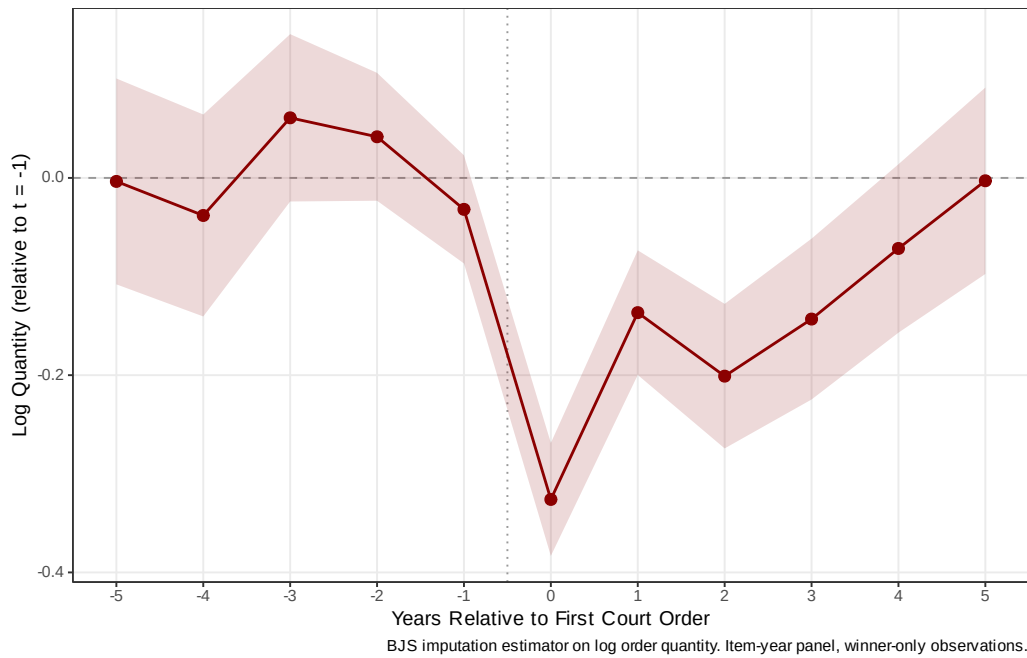


Figure A.16: Log Order Quantity Around First Court Order: Borusyak-Jaravel-Spiess Imputation

Notes: Item-year panel, winner-only observations. BJS imputation estimator on log order quantity. Within the same item, order quantity drops sharply at the year of the first court order (-27.8% at $t = 0$) and recovers thereafter (-0.3% at $t = +5$). The dynamic pattern documents the C1 fragmentation channel: judicial pressure forces smaller orders in real time, on the same item.

A.6 Heterogeneity

We examine heterogeneity in the urgency premium along four dimensions. Tables A.14–A.17 report results from split-sample regressions and interaction models using our preferred specification (item + year + PBU FE).

Table A.14: Heterogeneity: SUS Component

	Split Sample		Interaction
	Specialized (1)	Basic SUS (2)	Full Sample (3)
Urgent Purchase	0.005 (0.046)	0.030* (0.016)	-0.073 (0.097)
Urgent $\times$ Basic SUS			0.159 (0.115)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	45,048	151,835	196,883
Within R ²	0.000	0.000	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

SUS component (Table A.14)... The urgency premium on negotiated prices is concentrated among items in the basic SUS component (common medications): the coefficient is 0.030 ($p < 0.10$) for basic items and an insignificant 0.005 for specialized items, suggesting that procurement disruption is more consequential for items where established supply chains normally deliver better prices.

Time period (Table A.15)... The urgency premium is larger in the later period (2014–2019) than in the earlier period (2009–2013), consistent with the

Table A.15: Heterogeneity: Time Period

	Split Sample		Interaction
	Early (1)	Late (2014+) (2)	Full Sample (3)
Urgent Purchase	0.074*** (0.025)	0.045** (0.018)	0.259*** (0.035)
Urgent $\times$ Late Period			-0.339*** (0.043)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	82,658	113,936	196,883
Within R ²	0.000	0.000	0.005

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

growing volume and complexity of health litigation over time.

Market competition (Table A.16).. The urgency premium is three times larger in competitive markets (0.143 vs. 0.046), with a significant interaction (0.226, $p < 0.01$). In concentrated markets, fewer bidders leave less room for urgency to erode bargaining power; in competitive markets, the loss of planning time has the largest marginal impact.

PBU size (Table A.17).. The urgency premium is slightly larger for purchases made by smaller PBUs (below-median transaction volume), consistent with smaller units having less experience managing urgent procurement.

Table A.16: Heterogeneity: Market Competition

	Split Sample		Interaction
	Low Comp. (1)	High Comp. (2)	Full Sample (3)
Urgent Purchase	0.046*** (0.016)	0.143*** (0.034)	0.026 (0.018)
Urgent $\times$ High Competition			0.226*** (0.044)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	125,950	61,313	187,264
Within R ²	0.000	0.001	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.17: Heterogeneity: PBU Size

	Split Sample		Interaction
	Small PBU (1)	Large PBU (2)	Full Sample (3)
Urgent Purchase	0.023 (0.026)	0.062*** (0.018)	-0.000 (0.049)
Urgent $\times$ Large PBU			0.070 (0.054)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	31,902	164,511	196,883
Within R ²	0.000	0.000	0.000

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.7 Regex Classifier Validation

The purchase-type indicator is constructed by a regular-expression algorithm applied to public tender notices, which by Brazilian procurement law must reference any underlying court order. To document classifier accuracy, we draw a stratified random sample of 500 tender-notice subjects (approximately 167 per predicted class) and hand-label each as ordinary, administrative, or litigated. Hand-labeling of the stratified sample is in progress; F1 diagnostics will be reported in the replication archive accompanying this paper. The regex’s literal-reference design implies misclassification, if any, is near-symmetric across the three classes: the algorithm relies on textual markers that are present or absent independently of the price outcomes we estimate, so misclassification attenuates rather than biases our estimates.⁷

A.8 Selection Bounds, Reconciliation, and Diagnostic Robustness

This section collects the selection-bound, reconciliation, and diagnostic robustness exhibits that support the main-text claims in Sections 4 and 5.

Manski-Lee bounds on the UTG. Table A.18 reports the bounded UTG coefficient under monotone selection. Trimming admin observations within item×year×PBU strata where the admin sample exceeds the litigated count yields a bounded gap in [15.9%, 21.1%], against a naive coefficient of 29.5%.

⁷The stratified sample is reproduced from `output/validation/validation_sample.csv`, generated by `analysis/33_regex_validation_sample.R`; F1 statistics are emitted by `34_regex_validation_f1.R` when the labeled CSV is finalized.

Table A.18: Manski-Lee bounds on the under-the-gun coefficient.

Specification	log negotiated price		implied
	Coef. on Admin	SE	Lit-vs-Admin (%)
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530
Lee lower bound (admin top-trim)	-0.192	0.095	15.924
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123
Trimming applied within $item \times year \times PBU$ strata where the admin sample exceeds the li			

Average trimming intensity is 26.9% of admin observations across 12,038 of 44,654 strata.

Heckman selection model (sensitivity).. Table A.19 reports a parametric Heckman correction with the same first-stage probit (pre-period log reference price; pre-period mean quantity; pre-period mean number of bidders; pre-period order count—the exclusion restriction; SUS-formulary indicator). The first-stage probit on $N = 5,351$ observations confirms that pre-period reference price and SUS status both enter negatively and significantly. The second-stage Admin coefficient is 0.817 (SE 4.278); the inverse Mills ratio is collinear with the item, year, and PBU fixed effects, so the second-stage point estimate is uninformative. We treat this as a sensitivity diagnostic that flags identification fragility, not as a competing point estimate, and rely on the Lee bound as the primary selection-corrected object.

Both-types-cell representativeness.. Table A.20 compares the 5,581 item $\times$ year-month cells that contain both administrative and litigated urgent purchases (the within-cell variation that identifies the tightest specification) with the

Table A.19: Heckman selection-corrected UTG estimate.

Term	Estimate
First stage: pre-period log ref price	-0.051
First stage: pre-period mean qty	0.000
First stage: pre-period mean firms	0.065
First stage: pre-period n orders (excl)	0.000
First stage: SUS formulary	-0.265
Second stage: Admin coef	0.817
Second stage: inverse Mills ratio	-0.653
Second stage: implied lit-over-admin (%)	-55.841
Heckit ML: ρ	1.000

First stage: probit of admin on pre-period item characteristics. pre_n_orders is excluded from t

11,972 singleton cells. Both-types cells systematically have lower mean log reference and negotiated prices, larger quantities, slightly fewer bidders, and a higher SUS-formulary share. Bounded UTG estimates therefore speak to the more liquid, formulary-anchored items in the urgent panel; we read this as a generalizability boundary, not as a threat to within-cell identification.

Wild cluster bootstrap. Table A.21 reports Rademacher wild cluster bootstrap inference on the UTG coefficient (999 replicates, PBU clustering). The preferred-FE Admin coefficient is significant under both asymptotic and bootstrap inference ($p_{\text{boot}} = 0.0080$, 95% CI in log-points $[-0.469, -0.039]$). The item $\times$ year-month FE specification is borderline ($p_{\text{boot}} = 0.0390$), as expected given the thin within-cell sample.⁸

⁸Implementation: manual Rademacher implementation (the `fwildclusterboot` package is unavailable for R 4.5); see `analysis/44_wild_bootstrap.R`. Restricted-null residuals weighted by cluster-level Rademacher draws; bootstrap t -distribution constructed

Table A.20: Representativeness of both-types cells: cells with both administrative and litigated urgent purchases (5,581 cells, 31.8%) vs. singleton cells (11,972).

Variable	Both-types cells	Singleton cells
Log reference price	1.233	1.761
Log negotiated price	0.703	1.326
Log quantity	6.560	6.324
N. bidding firms	2.181	2.404
SUS formulary share	0.867	0.725
Reverse-auction (pregao) share	0.955	0.952

Means computed at the observation level within each cell type. p -values from two-sample t -tests.

Table A.21: Wild cluster bootstrap inference on the UTG coefficient.

Specification	$\hat{\beta}$	Asy. SE	t	p_{boot}	95% CI (log-points)
Preferred (item + year + PBU)	−0.259	0.092	−2.80	0.008	[−0.469, −0.039]
Tightest (item×year-month + PBU)	−0.283	0.116	−2.44	0.039	[−0.549, −0.001]

Rademacher wild cluster bootstrap on the Admin coefficient, clustered at the PBU level, with $B = 999$ replicates and restricted-null residuals. Negative coefficient means litigated purchases are more expensive than administrative ones, in log-points. Source: `analysis/44_wild_bootstrap.R`.

Rambachan-Roth honest sensitivity on BJS event-study. Figure A.17 reports the Borusyak-Jaravel-Spiess imputation event-study coefficients with a Rambachan-Roth honest sensitivity overlay. Pre-period maximum absolute coefficient is 0.041; the $t = 0$ effect is 0.052 (SE 0.018); the breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero is $M = 0.018$. The $t = 0$ effect is significant under parallel trends but does not survive the linear extrapolation of the observed pre-period maximum—we honor this in the main text by treating the cross-sectional

from 999 replicates.

fixed-effects estimate as primary and the BJS event study as supportive, not as a stand-alone identification claim.

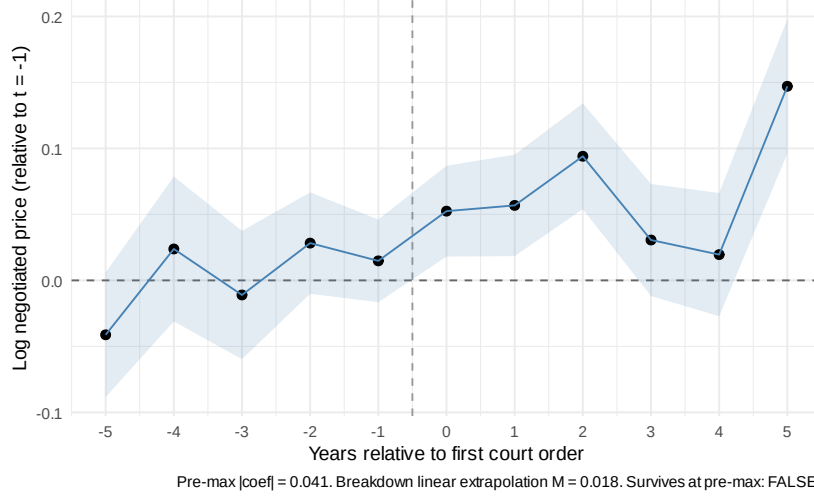


Figure A.17: BJS event study around first court order with Rambachan-Roth sensitivity diagnostic.

Reconciliation table.. Table A.22 decomposes the observed log-price gap (admin minus litigated) into the mechanical bulk-discount channel (-32.8% , admin cheaper from order-size differences alone), the within firm-buyer-item offset (3.5% , NULL), and the supplier composition residual (10.9%). Same firms price the same items the same way regardless of regime. The composition residual—positive in the admin-minus-lit direction—reflects that the litigated supplier mix is, conditional on quantity, less expensive per unit than the administrative supplier mix; the bulk-discount disadvantage of small lit orders dominates, producing the net lit-over-admin observed gap.

Table A.22: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log-points (admin minus lit)
Observed UTG (admin minus lit, log price)	-0.259
Mechanical C1: bulk-discount x qty gap	-0.397
Within firm-buyer-item offset	0.035
Supplier composition residual	0.104
Identity: observed UTG = mechanical C1 + within-firm offset + composition residual. Within-firm	

Welfare bound.. Table A.23 reports the selection-bias-corrected welfare bound.

Applying the Lee-midpoint UTG of 18.5% to \$300 M of annual litigated spending in São Paulo, with an admissibility-share calibration of 50.0%, yields \$27.8 M per year on the per-unit-price margin. The Lee-endpoint range is [\$23.9 M, \$31.7 M]. This is a per-unit-price bound and excludes welfare losses from constrained order sizes, reduced bidder participation, and officials' search time diverted from ordinary procurement; the headline number understates the total cost.

Table A.23: Selection-bias-corrected welfare bound on the procurement cost of judicial enforcement.

Component	Value
Bounded UTG (Lee midpoint)	18.5%
Lee lower bound	15.9%
Lee upper bound	21.1%
Admissibility share (calibration)	50%
Annual litigated spend, SP	\$300 M / yr
Welfare bound (Lee midpoint x admissibility x spend)	\$27.8 M / yr
Welfare bound (Lee lower x admissibility x spend)	\$23.9 M / yr
Welfare bound (Lee upper x admissibility x spend)	\$31.7 M / yr
Bounded UTG: Manski-Lee bounds from Table A.18 . Admissibility share: calibration anchor for	